

SAMARA SMYSHLYAEVKA, Russian Federation

WMO: **288070**

Lat: **53.250N** Lon: **50.450E** Elev: **131** StdP: **14.63** Time Zone: **4.00 (E04)** Period: **96-13** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB 99.6%	Heating DB 99%	Humidification 99.6% DP	Humidification 99.6% HR	Humidification 99.6% MCDB	Humidification 99% DP	Humidification 99% HR	Humidification 99% MCDB	Coldest Month 0.4% WS	Coldest Month 0.4% MCDB	Coldest Month 1% WS	Coldest Month 1% MCDB	MCWS/PCWD to 99.6% DB MCWS	MCWS/PCWD to 99.6% DB PCWD	WSF	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	**2**	**-19.4**	**-13.6**	**-24.6**	**1.4**	**-19.2**	**-18.7**	**2.0**	**-13.1**	**23.6**	**21.0**	**22.6**	**19.9**	**2.8**	**170**	**0.601**	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling 0.4% DB	Cooling 0.4% MCWB	Cooling 1% DB	Cooling 1% MCWB	Cooling 2% DB	Cooling 2% MCWB	Evaporation 0.4% WB	Evaporation 0.4% MCDB	Evaporation 1% WB	Evaporation 1% MCDB	Evaporation 2% WB	Evaporation 2% MCDB	MCWS/PCWD to 0.4% DB MCWS	MCWS/PCWD to 0.4% DB PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	**7**	**22.1**	**92.2**	**69.3**	**88.3**	**68.6**	**85.0**	**67.2**	**71.7**	**86.1**	**70.3**	**84.7**	**68.8**	**82.3**	**7.9**	**80**	(2)

	Dehumidification 0.4% DP	Dehumidification 0.4% HR	Dehumidification 0.4% MCDB	Dehumidification 1% DP	Dehumidification 1% HR	Dehumidification 1% MCDB	Dehumidification 2% DP	Dehumidification 2% HR	Dehumidification 2% MCDB	Enthalpy 0.4% Enth	Enthalpy 0.4% MCDB	Enthalpy 1% Enth	Enthalpy 1% MCDB	Enthalpy 2% Enth	Enthalpy 2% MCDB	Extreme Max WB	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	**67.0**	**100.0**	**77.8**	**65.5**	**94.5**	**75.9**	**63.9**	**89.5**	**74.6**	**35.5**	**86.1**	**34.3**	**84.6**	**33.0**	**82.2**	**78.6**	(3)

Extreme Annual Design Conditions

	Extreme Annual WS 1%	Extreme Annual WS 2.5%	Extreme Annual WS 5%		Extreme Annual Temperature Mean Min	Mean Max	Standard Deviation Min	Standard Deviation Max	n=5 years Min	n=5 years Max	n=10 years Min	n=10 years Max	n=20 years Min	n=20 years Max	n=50 years Min	n=50 years Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	**22.6**	**20.2**	**17.5**	DB	**-24.4**	**96.6**	**5.1**	**3.2**	**-28.1**	**98.9**	**-31.0**	**100.7**	**-33.9**	**102.5**	**-37.6**	**104.8**	(4)
(5)				WB	**-24.6**	**74.9**	**4.9**	**2.2**	**-28.1**	**76.5**	**-31.0**	**77.8**	**-33.8**	**79.0**	**-37.3**	**80.5**	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		**41.8**	**13.1**	**12.4**	**25.1**	**44.4**	**57.3**	**65.9**	**70.8**	**66.9**	**55.3**	**43.0**	**28.7**	**16.7**	(6)
(7)		DBStd		**23.12**	**13.41**	**12.59**	**9.38**	**10.07**	**8.36**	**7.80**	**6.42**	**7.33**	**7.12**	**7.75**	**10.98**	**14.03**	(7)
(8)		HDD50		**5167**	**1143**	**1053**	**773**	**223**	**30**	**2**	**0**	**0**	**26**	**245**	**640**	**1032**	(8)
(9)		HDD65		**8937**	**1608**	**1473**	**1238**	**619**	**260**	**82**	**22**	**66**	**300**	**683**	**1089**	**1497**	(9)
(10)		CDD50		**2173**	**0**	**0**	**0**	**56**	**258**	**478**	**645**	**523**	**186**	**27**	**0**	**0**	(10)
(11)		CDD65		**469**	**0**	**0**	**0**	**2**	**23**	**108**	**202**	**124**	**10**	**0**	**0**	**0**	(11)
(12)		CDH74		**5507**	**0**	**0**	**0**	**35**	**379**	**1214**	**2263**	**1435**	**181**	**0**	**0**	**0**	(12)
(13)		CDH80		**2110**	**0**	**0**	**0**	**3**	**92**	**441**	**935**	**604**	**35**	**0**	**0**	**0**	(13)
(14)	Wind	WSAvg		**7.2**	**7.2**	**7.3**	**8.0**	**8.1**	**8.2**	**7.0**	**6.1**	**6.2**	**6.1**	**7.2**	**7.9**	**7.4**	(14)
(15)	Precipitation	PrecAvg		**20.9**	**1.8**	**1.3**	**1.2**	**1.5**	**1.1**	**2.3**	**2.3**	**1.9**	**1.7**	**2.0**	**1.9**	**1.9**	(15)
(16)		PrecMax		**25.5**	**3.6**	**3.6**	**3.4**	**3.9**	**2.6**	**6.6**	**4.6**	**4.3**	**4.1**	**4.2**	**5.9**	**5.2**	(16)
(17)		PrecMin		**11.2**	**0.6**	**0.0**	**0.1**	**0.0**	**0.0**	**0.0**	**0.3**	**0.0**	**0.0**	**0.5**	**0.3**	**0.4**	(17)
(18)		PrecStd		**3.0**	**0.8**	**0.9**	**0.7**	**1.0**	**0.7**	**1.5**	**1.4**	**1.2**	**1.1**	**1.0**	**1.2**	**1.0**	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		**36.9**	**36.4**	**53.1**	**78.3**	**86.9**	**95.4**	**97.4**	**97.3**	**84.3**	**70.0**	**53.1**	**40.1**	(19)
(20)			MCWB		**34.9**	**33.9**	**45.2**	**57.9**	**63.7**	**68.2**	**69.1**	**69.0**	**61.4**	**54.7**	**50.4**	**38.3**	(20)
(21)		2%	DB		**35.0**	**35.0**	**44.4**	**72.5**	**82.4**	**90.0**	**91.9**	**91.2**	**79.4**	**64.7**	**47.6**	**36.7**	(21)
(22)			MCWB		**33.5**	**33.3**	**38.7**	**55.1**	**62.4**	**67.9**	**70.1**	**68.4**	**61.0**	**52.3**	**44.6**	**34.7**	(22)
(23)		5%	DB		**33.6**	**33.7**	**39.4**	**67.7**	**78.3**	**85.0**	**88.5**	**86.8**	**74.3**	**59.6**	**44.3**	**35.1**	(23)
(24)			MCWB		**32.5**	**32.5**	**35.6**	**52.8**	**60.1**	**67.2**	**69.6**	**67.8**	**59.6**	**50.5**	**42.1**	**33.5**	(24)
(25)		10%	DB		**31.5**	**31.3**	**36.7**	**62.1**	**73.9**	**80.8**	**85.5**	**82.1**	**69.2**	**55.6**	**41.3**	**33.7**	(25)
(26)			MCWB		**30.4**	**30.0**	**33.9**	**50.0**	**58.4**	**65.5**	**68.4**	**65.9**	**57.4**	**49.1**	**39.4**	**32.5**	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		**35.6**	**34.5**	**45.4**	**60.0**	**67.4**	**72.0**	**74.6**	**72.3**	**65.9**	**57.5**	**51.4**	**39.1**	(27)
(28)			MCDB		**36.6**	**35.6**	**53.2**	**74.7**	**81.5**	**87.6**	**87.7**	**88.9**	**76.7**	**63.1**	**53.1**	**40.0**	(28)
(29)		2%	WB		**33.9**	**33.5**	**38.9**	**57.1**	**64.3**	**70.2**	**72.6**	**70.5**	**62.9**	**54.3**	**45.4**	**35.2**	(29)
(30)			MCDB		**34.7**	**34.7**	**43.7**	**69.9**	**78.2**	**85.3**	**86.0**	**87.0**	**74.4**	**61.5**	**47.3**	**36.3**	(30)
(31)		5%	WB		**32.6**	**32.5**	**36.0**	**53.6**	**62.1**	**68.5**	**71.0**	**68.9**	**60.9**	**51.7**	**42.3**	**33.7**	(31)
(32)			MCDB		**33.7**	**33.7**	**38.6**	**65.5**	**75.9**	**81.7**	**84.2**	**84.4**	**71.7**	**58.2**	**44.1**	**34.6**	(32)
(33)		10%	WB		**30.3**	**29.9**	**34.2**	**50.5**	**59.6**	**66.8**	**69.8**	**66.8**	**58.5**	**49.4**	**39.6**	**32.5**	(33)
(34)			MCDB		**31.5**	**31.3**	**36.4**	**61.3**	**70.8**	**78.6**	**82.9**	**79.6**	**67.9**	**55.1**	**41.2**	**33.7**	(34)
(35)	Mean Daily Temperature Range		MDBR		**12.3**	**15.0**	**15.1**	**19.3**	**22.4**	**21.6**	**22.1**	**22.7**	**21.3**	**15.8**	**9.5**	**10.8**	(35)
(36)		5% DB	MCDBR		**9.9**	**9.6**	**15.3**	**29.0**	**29.5**	**27.4**	**27.6**	**29.8**	**30.5**	**24.9**	**11.5**	**7.5**	(36)
(37)			MCWBR		**9.3**	**8.6**	**10.6**	**14.4**	**13.4**	**11.3**	**10.8**	**11.8**	**14.7**	**14.3**	**9.8**	**7.2**	(37)
(38)		5% WB	MCDBR		**9.9**	**8.9**	**14.1**	**26.2**	**26.9**	**23.5**	**24.0**	**26.9**	**26.2**	**19.9**	**10.3**	**7.0**	(38)
(39)			MCWBR		**9.5**	**8.7**	**10.4**	**13.9**	**13.3**	**10.8**	**10.4**	**11.7**	**13.9**	**12.5**	**9.4**	**6.9**	(39)
(40)	Clear-Sky Solar Irradiance	taub			**0.305**	**0.328**	**0.375**	**0.432**	**0.408**	**0.401**	**0.411**	**0.430**	**0.385**	**0.355**	**0.331**	**0.303**	(40)
(41)		taud			**2.063**	**2.031**	**1.995**	**2.086**	**2.271**	**2.343**	**2.312**	**2.240**	**2.338**	**2.332**	**2.235**	**2.085**	(41)
(42)		Ebn at Noon			**198**	**234**	**249**	**251**	**267**	**270**	**264**	**251**	**248**	**229**	**193**	**173**	(42)
(43)		Edh at Noon			**24**	**34**	**43**	**45**	**40**	**37**	**38**	**39**	**31**	**25**	**20**	**19**	(43)
(44)	All-Sky Solar Radiation	RadAvg			**292**	**630**	**1060**	**1376**	**1789**	**1927**	**1905**	**1559**	**1031**	**520**	**222**	**162**	(44)
(45)		RadStd			**46**	**63**	**122**	**142**	**133**	**212**	**161**	**128**	**145**	**90**	**37**	**45**	(45)

Historical Trends

		DBAvg	Heating 99% DB	Heating 99% DP	Cooling 1% DB	Cooling 1% WB	Cooling 1% DP	Degree-Days HDD50	Degree-Days HDD65	Degree-Days CDD50	Degree-Days CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(46)
(47)	Regional (5 neighbors)	**N/A**	**N/A**	**N/A**	**+1.96**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	**N/A**	(47)

Nomenclature: See separate page